



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Kinetic Model of Matter

2.1

STUDENT NAME



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PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

2.1 Kinetic particle model of matter

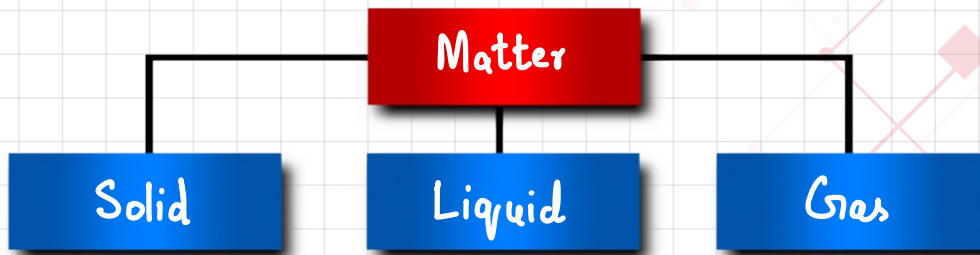
2.1.1 States of matter

- 1 Know the distinguishing properties of solids, liquids and gases
- 2 Know the terms for the changes in state between solids, liquids and gases (gas to solid and solid to gas transfers are **not** required)

2.1.2 Particle model

- 1 Describe, qualitatively, the particle structure of solids, liquids and gases, relating their properties to the forces and distances between particles and to the motion of the particles (atoms, molecules, ions and electrons)
- 2 Describe the relationship between the motion of particles and temperature, including the idea that there is a lowest possible temperature (-273°C), known as absolute zero, where the particles have least kinetic energy
- 3 Describe the pressure and the changes in pressure of a gas in terms of the forces exerted by particles colliding with surfaces, creating a force per unit area
- 4 Explain qualitatively, in terms of particles, the relationship between:
 - (a) pressure and temperature at constant volume
 - (b) volume and temperature at constant pressure
 - (c) pressure and volume at constant temperature
- 5 Recall and use the equation $p_1V_1 = p_2V_2$, including a graphical representation of the relationship between pressure and volume for a gas at constant temperature

STATES OF MATTER:



SOLID: "A state of matter which has definite shape, high density & particles are tightly packed & vibrate only."

LIQUID: "A state of matter which has indefinite shape, medium density & particles are closer with each other & move randomly at high speeds in clusters."

GAS: "A state of matter which has indefinite shape, low density & particles are away from each other & move randomly freely."

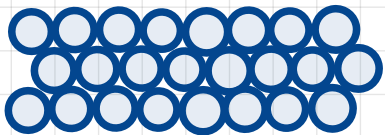
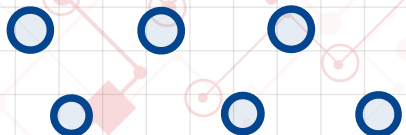
STATE CHANGES.

MELTING: Conversion of state from solid to liquid

FREEZING: Conversion of state from liquid to solid

VAPORIZATION: Conversion of state from liquid to gas

CONDENSATION: Conversion of state from gas to liquid

SOLIDS	LIQUIDS	GASES
Particles of solids are closely packed. 	Particles of liquid are arranged in the form of clusters. 	Particles of gases are free to move everywhere. 
Intermolecular separation is very low.	Inter molecular separation is low but higher than solids.	Inter molecular separation is very high.
Intermolecular forces are very strong.	Intermolecular forces are stronger but weaker than solids.	Intermolecular forces are very weak.
They have high density.	They have moderate density.	They have low density.
Molecules vibrate about their mean position.	Molecules move past each other with high speeds.	Molecules are free to move with high speeds.
They have fixed shape.	They don't have fixed shape.	They don't have fixed shape.
They have fixed volume.	They have fixed volume.	They don't have fixed volume.
Incompressible	Incompressible	Compressible

TEMPERATURE & KINETIC ENERGY.

In solids, at room temperature, the particles vibrate about fixed positions.

- When temperature increases, vibration of particles increases hence kinetic energy increases.
- When temperature decreases, vibration of particles decreases hence kinetic energy decreases eventually to zero at temperature -273°C .

This temperature is called Absolute Zero.

PRESSURE & KINETIC ENERGY:

GAS PRESSURE:

Particles of gas move rapidly in random directions with wide range of speeds thus collide with the walls of the container repeatedly. So the gas molecules exert force & eventually pressure on the container.

When temperature increases, average speed of molecules & kinetic energy increases so now particles hit more frequently hence pressure increases.

FORCE & GAS PRESSURE:

At constant temperature, the average force & hence pressure is constant

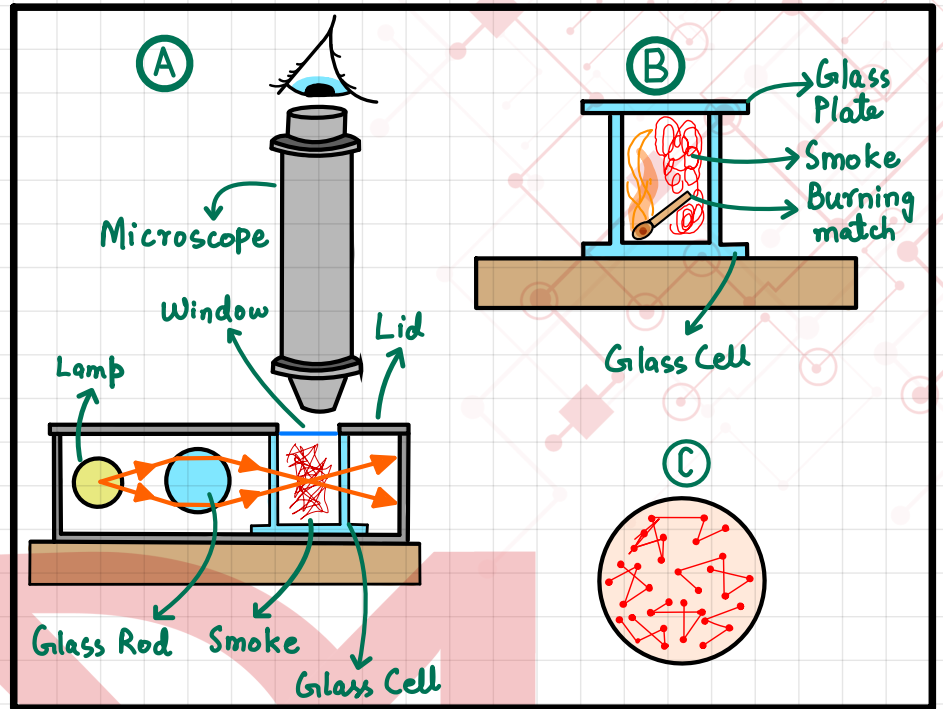
- As temperature increases, average force & pressure also increases.

BROWNIAN MOTION:

"The movement of air particles in haphazardous random motion in a straight line is called Brownian motion."

RANDOMNESS:

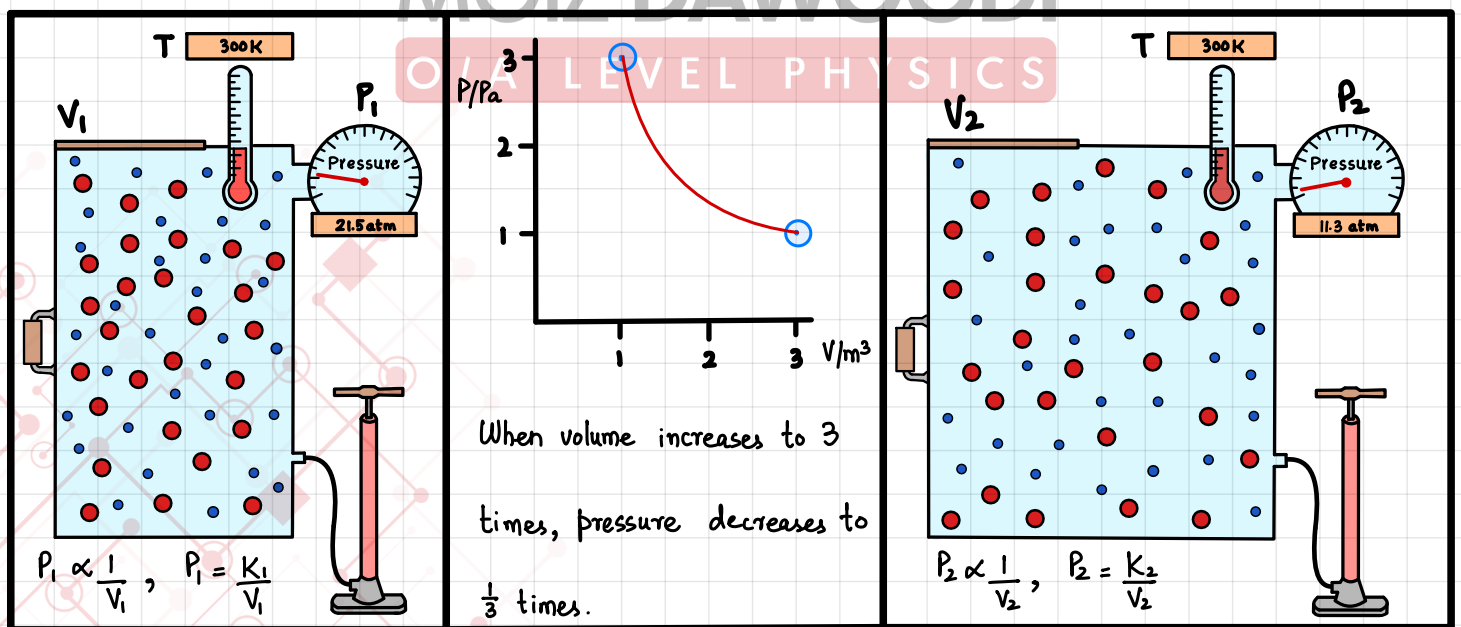
Randomness of molecules increases with increase in temperature.



GAS RELATIONSHIPS:

VOLUME - PRESSURE RELATIONSHIP (BOYLE'S LAW):

"Volume is inversely proportional to the pressure provided that temperature & no. of molecules remains constant."

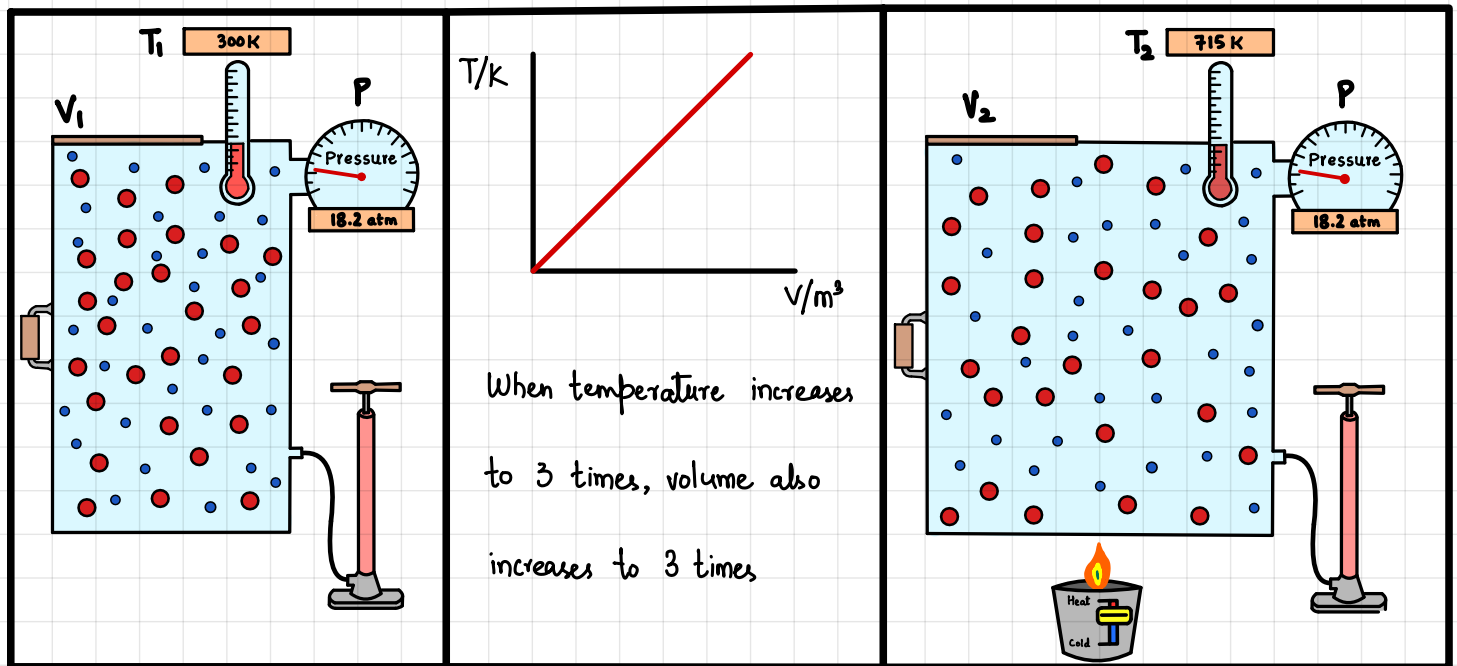


So, $K_1 = K_2$

$$P_1 V_1 = P_2 V_2$$

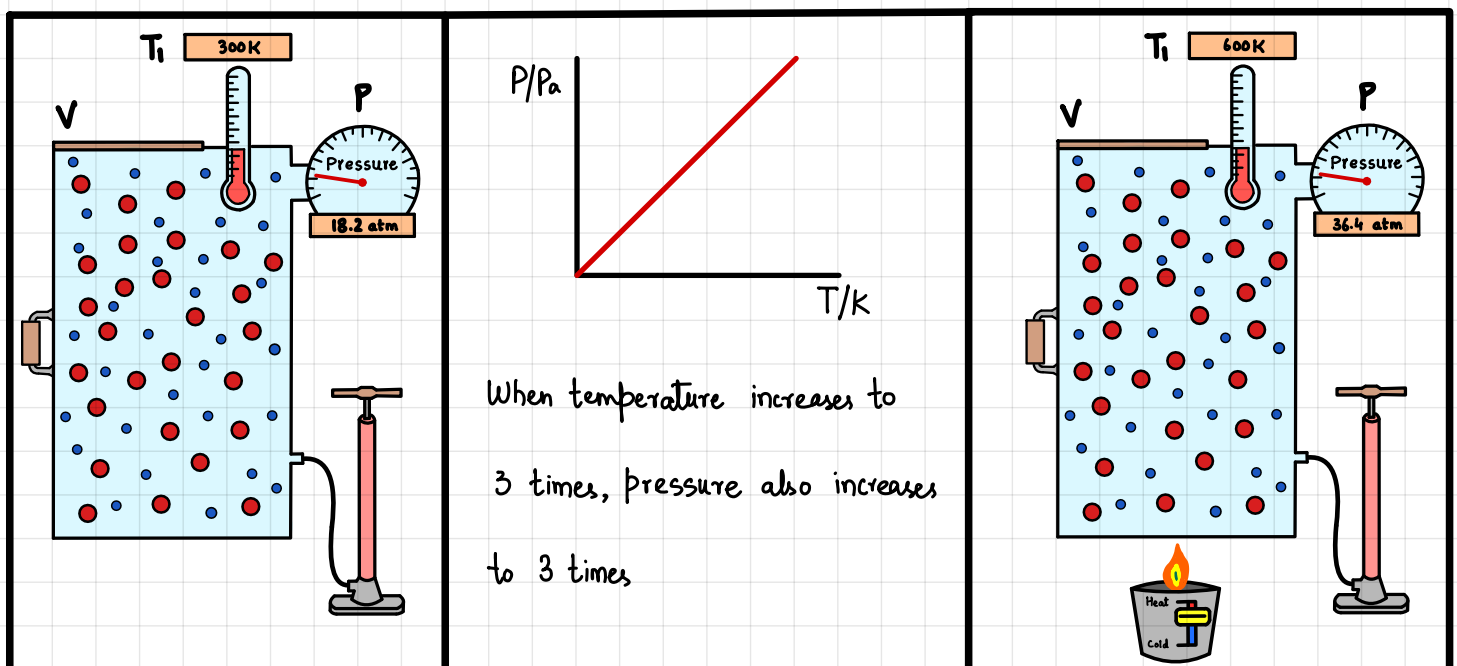
● VOLUME - TEMPERATURE RELATIONSHIP.

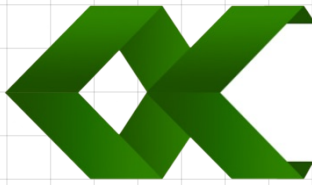
"Volume is directly proportional to the temperature provided that pressure & no. of molecules remains constant."



● PRESSURE - TEMPERATURE RELATIONSHIP.

"Pressure is directly proportional to the temperature provided that volume & no. of molecules remains constant."





MATCH THE FOLLOWING

In melting,	vibrate about their fixed positions
In boiling,	solid changes its state to liquid.
Particles of solid,	Liquid changes state to gas.
Liquid molecules have	are free to move
Molecules of gas	compressibles.
Gases are considered as	strong intermolecular forces.
Volume is inversely proportional	proportional to the temperature at constant pressure.
Volume is directly	to the pressure at constant temperature.
Pressure is directly	density & strong intermolecular forces.
Solids have high	proportional to the temperature at constant volume.
Liquids have	indefinite shape & fixed volume.

01 According to the kinetic theory, matter is made up of very small particles in a constant state of motion.

Which row best describes the particle behaviour in the liquid state?

	forces between particles	motion of particles
A	strong ✓	move randomly at high speed ✓
B	strong ✓	vibrate but are free to move position ✗
C	strong	vibrate to and fro around a fixed position ✗
D	weak	move randomly at high speed ✓

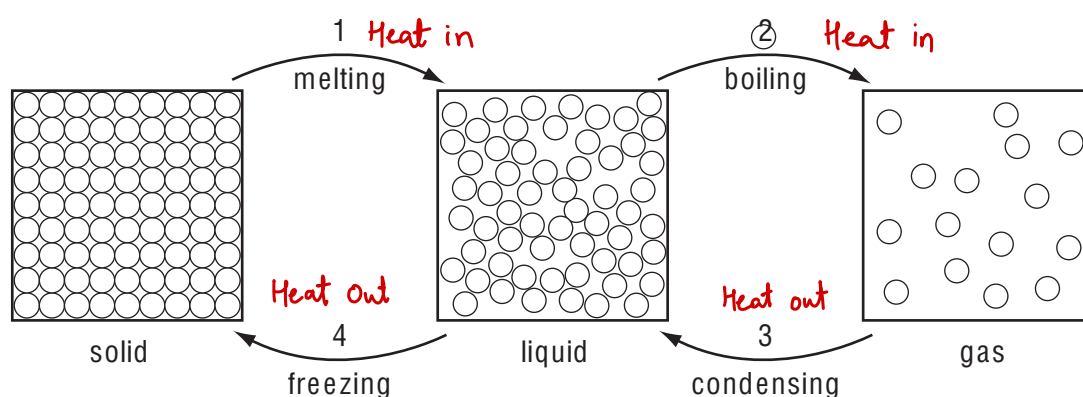
[MJ2011/P11/Q15]

02 Which row explains why a liquid has a fixed volume but does not have a fixed shape?

	force between molecules in the liquid	movement of molecules in the liquid
A	large ✓	move throughout the liquid ✓
B	large ✓	vibrate at fixed positions
C	small	move throughout the liquid ✓
D	small	vibrate at fixed positions

[ON2011/P11/Q15]

03 Substances can change from one state to another as shown.



For substances to change from one state to another, there must be some energy transfer.

Which changes involve the substance taking in energy and which changes involve the substance giving out energy?

	energy taken in	energy given out
A	1 and 2 ✓	3 and 4 ✓
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

[ON2011/P11/Q19]

$$P \propto T$$

- 04 A gas is enclosed in a container of fixed volume. It gains heat energy from an external source. What happens to the molecules of the gas?

- ☒ A They expand.
☒ B They move faster inside the container. ✓
☒ C They move further apart.
☒ D They vibrate with greater frequency.

[MJ2012/P11/Q15]

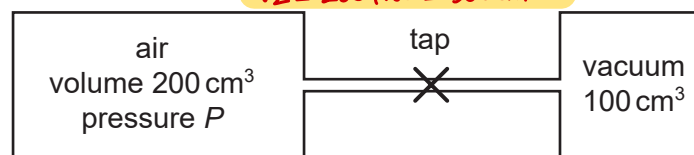
- 05 Two cylinders are connected by a thin pipe.

One cylinder has a volume of 200 cm^3 and contains air at pressure P .

$$V_1 \quad P_1$$

The other cylinder has a volume of 100 cm^3 and contains a vacuum (no air).

$$V_2 = 200 + 100 = 300 \text{ cm}^3$$



Initially, the tap is closed.

$$P_2 = ?$$

What is the final pressure of the air after the tap is opened?

- A $\frac{P}{2}$ ☒ B $\frac{2P}{3}$ C $\frac{3P}{2}$ D $2P$

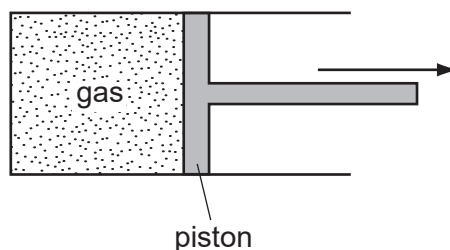
$$P_1 V_1 = P_2 V_2$$

$$\frac{P_1 V_1}{V_2} = P_2$$

$$\frac{P \times 200}{300} = P_2$$

[MJ2012/P12/Q12]

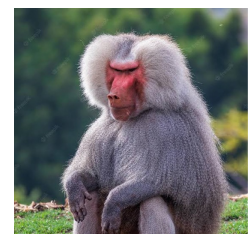
- 06 A fixed mass of gas is enclosed in a cylinder by a movable piston.



The piston is moved so that the volume occupied by the gas increases. The temperature remains constant.

What happens to the pressure of the gas and why does this happen?

	pressure	reason
A	decreases ✓	the molecules move more slowly
☒ B	decreases ✓	the molecules collide with the piston less frequently ✓
C	increases	the molecules move more quickly
D	increases	the molecules collide with the piston more frequently ✓



[MJ2012/P12/Q15]

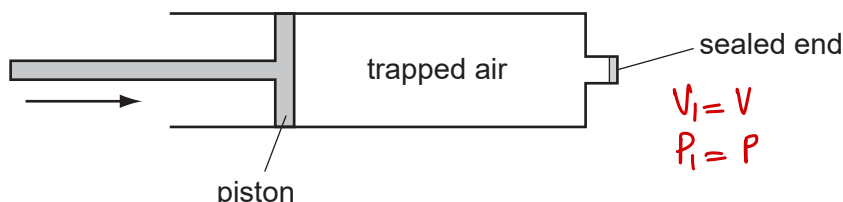
07 When a gas is rapidly compressed to a smaller volume, its temperature increases.

What happens to the gas molecules?

- A They move closer together and their average speed decreases. ✗
- B They move closer together and their average speed increases. ✓
- C They move closer together and their average speed remains unchanged. ✗
- D They stay the same distance apart and their average speed increases. ✗

[ON2012/P11/Q15]

08 The end of a bicycle pump is sealed.



$$V_1 = V$$
$$P_1 = P$$

$$V_2 = \frac{1}{4} V$$
$$P_2 = 4P$$

The temperature of the air does not change as the piston is pushed in. The final volume of trapped air is $\frac{1}{4}$ of the original volume.

How does the pressure of the trapped air change?

- A It decreases to $\frac{1}{4}$ of the original value.
- B It decreases to $\frac{3}{4}$ of the original value.
- C It increases to 3 times the original value.
- D It increases to 4 times the original value. ✓

[ON2012/P12/Q12]

09 A gas in a container of fixed volume is heated.

What happens to the molecules of the gas?

- A They collide less frequently. ✗
- B They expand. ✗
- C They move faster. ✓
- D They move further apart. ✗

[ON2012/P12/Q15]

10 When a gas in a container of fixed volume is heated, the pressure of the gas increases.

Which statement explains this?

- A The molecules expand and get heavier, so they hit each other harder. ✗
- B The molecules have less room to move, so they collide with each other more frequently. ✗
- C The molecules hit the walls harder and less frequently. ✗
- D The molecules move faster, so they collide with the walls more frequently. ✓

[MJ2013/P11/Q16]

- 11 An airtight container holds a fixed mass of gas. Its pressure and volume are measured on four occasions when the temperature is 20 °C.

The results are shown in the table. Three sets of readings are correct.

Which set of readings is **not** correct?

	pressure / kPa		volume / cm ³
A	120	X	36 = 4320
B	100	X	48 = 4800
C	80	X	60 = 4800
D	60	X	80 = 4800

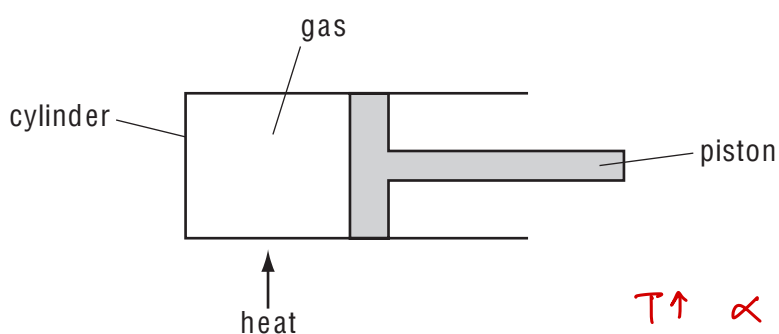
$$V \propto \frac{1}{P}$$

$$V = \frac{K}{P}$$

$$PV = K$$

[MJ2013/P12/Q12]

- 12 The diagram shows a fixed mass of gas in a cylinder fitted with a piston that can move easily.



$$T \uparrow \propto V \uparrow$$

What is the change, if any, in the pressure and volume of the gas after it is heated?

	pressure	volume
A	no change ✓	increases ✓
B	decreases	no change
C	decreases	increases ✓
D	increases	decreases

[MJ2013/P12/Q20]

- 13 Which row correctly describes the shape and volume of a gas?

	shape	volume
A	fixed	fixed
B	fixed	not fixed ✓
C	not fixed ✓	fixed
D	not fixed ✓	not fixed ✓

[ON2013/P11/Q18]

14 The conditions at the **bottom** and at the **surface of a lake** are given in the table.

100k

500k

	bottom of lake	surface of lake
temperature	10 °C	10 °C
pressure	P_1 500 kPa	P_2 100 kPa

A bubble of volume V_1 **1.0 cm³** forms at the bottom of the lake.

What is the volume of the bubble as it reaches the surface?

- A** 0.20 cm³ **B** 0.25 cm³ **C** 4.0 cm³ **D** 5.0 cm³

Boyle's Law

$$P_1 V_1 = P_2 V_2$$

$$500k \times 1 = 100k V_2$$

$$V_2 = \frac{500k}{100k}$$

$$V_2 = 5 \text{ cm}^3$$

[ON2013/P12/Q12]

15 A **fixed mass** of **gas** is kept at **constant pressure**. Its **temperature is raised**.

What happens to the **volume** of the gas and to its **molecules**?

	volume	molecules
A	decreases	move more slowly
B	increases ✓	stay the same distance apart
C	increases ✓	move further apart ✓
D	no change	move at the same speed

$$\uparrow V \propto T \uparrow$$

[ON2013/P12/Q19]

16 Air is heated in a sealed container with constant volume.

Why does the air pressure increase when the temperature increases?

- A** The air molecules expand.
B The air molecules bounce off each other more frequently.
C The air molecules bounce off the walls more frequently.
D The number of air molecules increases.

[ON2013/P12/Q20]

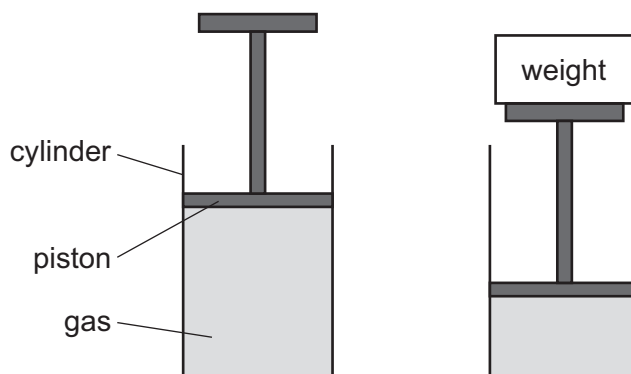
17 A gas occupies a volume of 2.0 m³ in a cylinder at a pressure of 240 kPa. A piston compresses the gas until the volume is 0.50 m³, the temperature remaining constant.

What is the new pressure of the gas?

- A** 60 kPa **B** 240 kPa **C** 480 kPa **D** 960 kPa

[MJ2014/P11/Q14]

- 18 A piston is supported by gas trapped in a cylinder.



A weight is put on the piston. The volume of gas supporting the piston decreases but the temperature of the gas is unchanged.

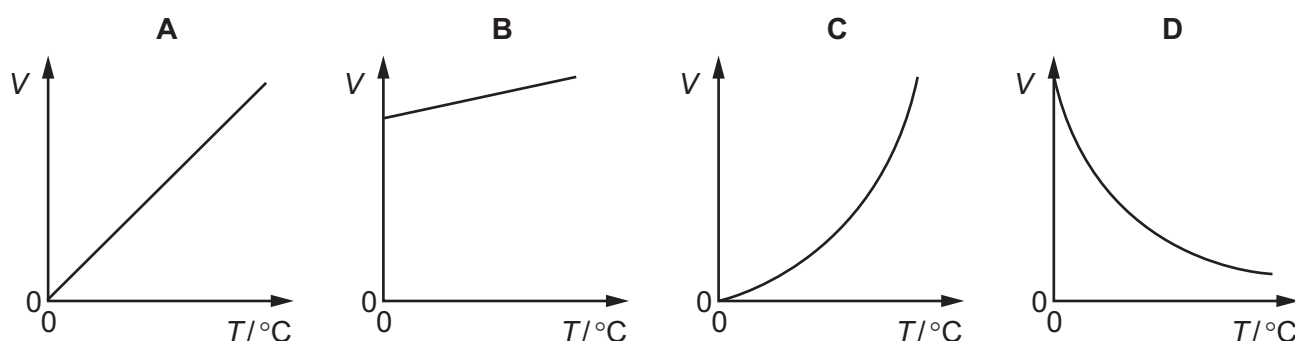
What happens to the molecules?

- A They hit the piston more frequently.
- B They move more slowly.
- C They have a smaller size.
- D They have more kinetic energy.

[MJ2014/P11/Q18]

- 19 A fixed mass of gas in a syringe at 0°C is heated at constant pressure.

Which graph shows the variation of volume V with temperature T , measured in $^\circ\text{C}$?

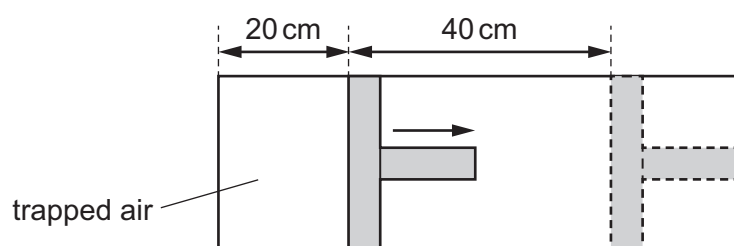


[MJ2014/P11/Q21]

- 20 Air is trapped in a cylinder by a piston. The pressure of the air is p and the length of the air column is 20 cm.

The piston is moved outwards until the length of the air column has increased by 40 cm.

The temperature of the air remains constant.



What is the new air pressure?

- A $\frac{p}{2}$
- B $\frac{p}{3}$
- C $2p$
- D $3p$

[ON2014/P11/Q12]

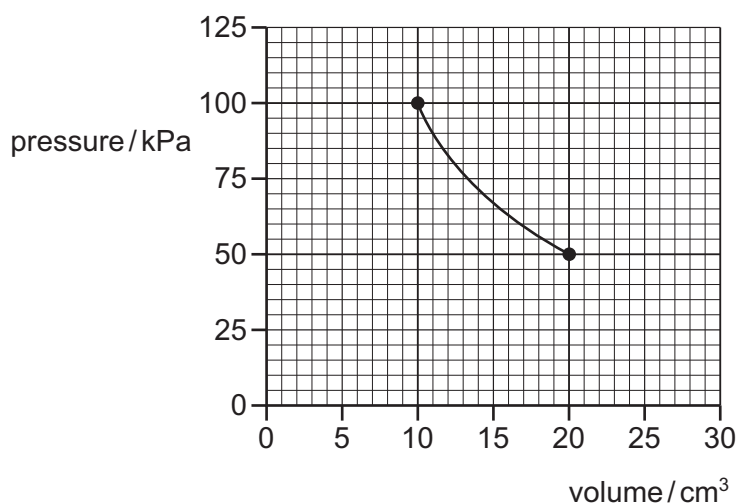
21 At a constant temperature, a solid has a fixed shape and a fixed volume.

Which row describes the shape and the volume of a liquid at constant temperature?

	shape	volume
A	fixed	fixed
B	fixed	not fixed
C	not fixed	fixed
D	not fixed	not fixed

[ON2014/P12/Q15]

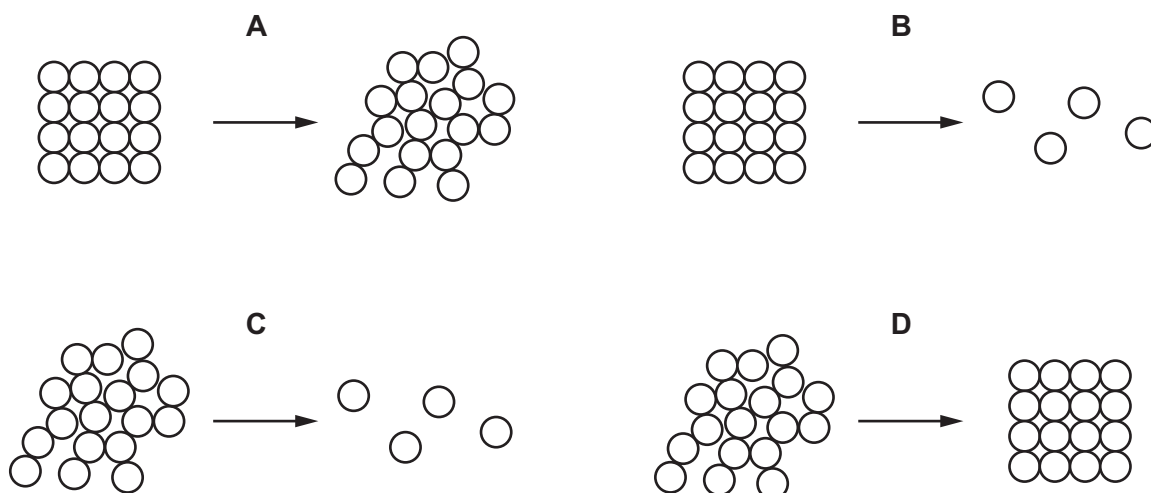
22 The graph shows how the pressure of a fixed mass of gas varies with volume at constant temperature.



What is the volume of the gas when the pressure is 25 kPa?

A 2.5 cm³ **B** 10 cm³ **C** 30 cm³ **D** 40 cm³ [MJ2015/P12/Q11]

23 Which diagram represents the change in the arrangement of the molecules in a solid as the substance melts?



[MJ2015/P12/Q19]

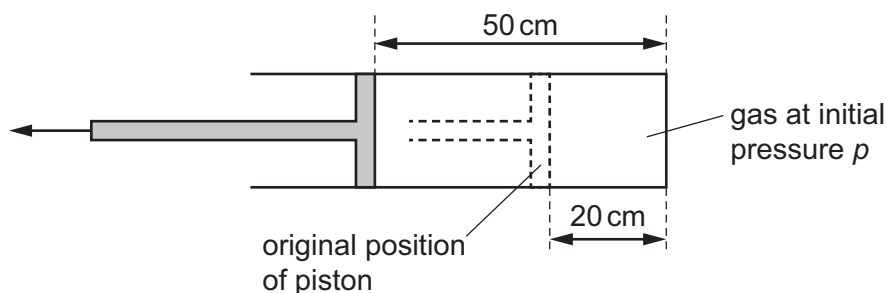
- 24 Some gas is trapped in a closed container. The gas is cooled and the volume of the container is kept constant.

What happens to the gas molecules?

- A They collide with the walls more often.
- B They contract.
- C They get closer together.
- D They move more slowly.

[MJ2015/P12/Q20]

- 25 A gas is trapped inside a cylinder by a movable piston. The length of the gas column is 20 cm and the pressure inside the cylinder is p .



The piston is pulled out a distance of 30 cm, so that the length of the gas column is now 50 cm. The temperature of the gas does not change.

What is the new pressure of the gas?

- A $0.40p$ B $0.60p$ C $1.5p$ D $2.5p$ [ON2015/P11/Q13]

- 26 A gas is in a sealed container of constant volume. The gas is heated and the pressure of the gas on the walls of the container increases.

How do the particles of the gas cause this increase in pressure?

- A They expand.
- B They hit each other more frequently.
- C They hit the container more frequently.
- D They vibrate faster.

[ON2015/P12/Q16]

ANSWER KEY				
1	B		21	C
2	A		22	A
3	A		23	A
4	B		24	D
5	B		25	A
6	B		26	C
7	B			
8	D			
9	C			
10	D			
11	A			
12	A			
13	D			
14	D			
15	C			
16	C			
17	D			
18	A			
19	B			
20	B			

- 1 A student notices puddles of water on a road, as shown in Fig. 1.1.



Fig. 1.1

Later in the day, he passes the puddles again and some of the water has evaporated.

- (a) State two changes to the atmospheric conditions that would cause the water to evaporate faster.

1.

2.

[2]

- (b) Explain, in terms of molecules, what happens during evaporation.

.....

.....

.....

.....

.....[2]

2 (a) Explain, using ideas about molecules,

(i) why a balloon filled with gas expands when heated,

As temperature increases, speed of molecules increases too this increases the collision of molecules hence force on a given area increases so balloon expands provided that no. of molecules are kept constant. [2]

(ii) why a balloon filled with water expands very little when heated.

As liquids have strong intermolecular forces so they expand less. [1]

(b) (i) A bubble of gas rises from the bottom of a lake to the surface. The pressure at the bottom of the lake is $3.0 \times 10^5 \text{ Pa}$ and the pressure at the surface is $1.0 \times 10^5 \text{ Pa}$. The volume of the bubble at the bottom of the lake is 2.0 cm^3 .

Atmospheric Pressure + Hydrostatic Pressure

P_2 Atmospheric Pressure

Calculate the volume of the bubble at the surface. $V_2 = ?$

$$P_1 V_1 = P_2 V_2$$

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{3 \times 10^5 \times 2}{1 \times 10^5}$$

volume = 6 cm^3 [2]

(ii) State one assumption that you have made in your calculation in (i).

Temperature / No. of molecules remains constant. [1]

[MJ2011/P22/Q3]

- 3 A helium balloon carries scientific instruments high up in the atmosphere.

Fig. 3.1 shows the partially inflated helium balloon leaving the ground.

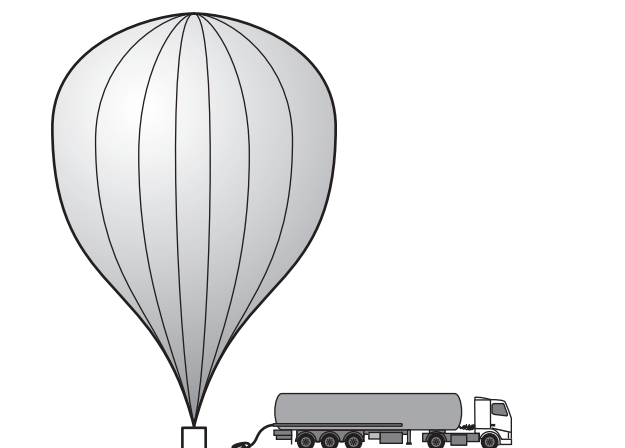


Fig. 3.1

- (a) A completely deflated balloon contains no gas. The helium to inflate it is stored in a very large cylinder at a pressure p_1 of 2.5×10^7 Pa. Helium that occupies a volume V_1 of 18 m^3 in the cylinder is slowly released into the balloon until the pressure p_2 in the balloon is 1.0×10^5 Pa. The temperature of the helium remains constant.

- (i) State the equation that relates the volume V_2 of the helium in the balloon at launch to p_1 , p_2 and V_1 .

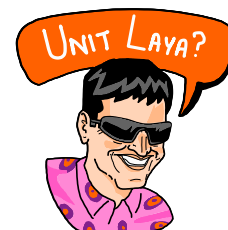
$$p_1 V_1 = p_2 V_2 \dots\dots\dots [1]$$

- (ii) Calculate V_2 .

$$p_1 V_1 = p_2 V_2$$

$$V_2 = \frac{p_1 V_1}{p_2} = \frac{2.5 \times 10^7 \times 18}{1 \times 10^5}$$

$$V_2 = 4500 \text{ m}^3 \dots\dots\dots [2]$$



- (b) When it leaves the ground, the balloon is only partially inflated. Suggest and explain why this is necessary.

Pressure is low at height so balloon may get exploded as molecules of balloon try to move from balloon to outside.

..... [2]

- 4 An experiment is carried out to find how the pressure of a fixed mass of air at room temperature varies with volume. Fig. 4.1 shows the apparatus used. The syringe is sealed at one end and the piston is free to move up and down as different metal weights are used.

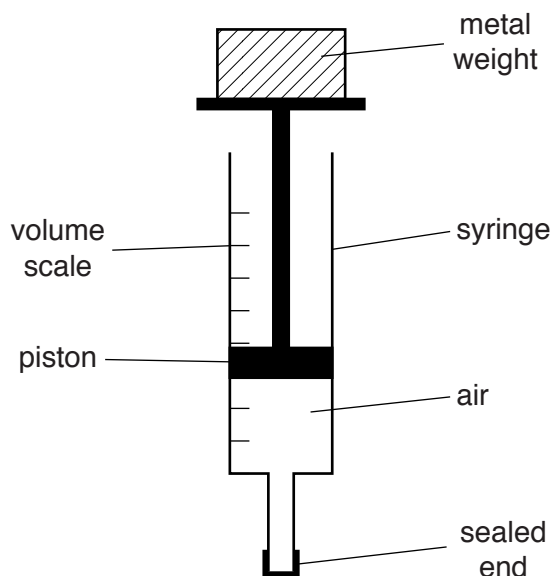


Fig. 4.1

- (a) State the unit in which pressure is measured.

Pascals (Pa)

[1]

- (b) Fig. 4.2 shows the axes for a graph of pressure against volume for the air in the syringe.

One point is plotted on the graph at pressure of P_0 and volume V_0 .

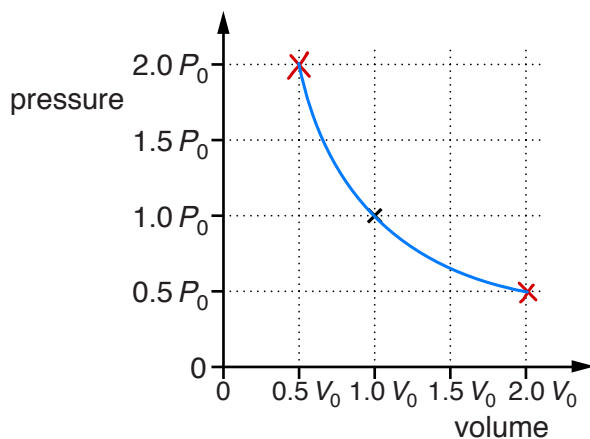


Fig. 4.2

The temperature of the air is kept constant.

On Fig. 4.2,

- (i) plot points at volumes of $0.5 V_0$ and $2.0 V_0$,

[1]

- (ii) complete the graph.

[1]

(c) More metal weights are placed on top of the syringe.

Explain how the molecules of air inside the syringe are able to support more metal weights.

More weight decreases volume of gas.

This increases pressure of gas

Since pressure is force per unit area so upward force increases

[3]

[MJ2015/P21/Q3]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2010	5054	22

- 4 (a) increased/high(er) temperature/hot(ter)
wind or air flow
less humidity
less pressure ANY 2 lines B2 [2]
- (b) molecules/atoms/particles escape/leave (surface)
or molecules become gaseous/vapour C1
or molecules break bonds
molecules with large(est) energy/high(est) speed sufficient or enough energy
escape/break bonds/become gaseous or leave behind slow/less energetic
molecules A1 [2]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – May/June 2011	5054	22

- 3 (a) (i) molecules have more **kinetic** energy/speed/velocity B1
hit sides hard(er)/with more force // (initially) hit sides (more) often/frequently
// create large(r) pressure (initially) B1
- (ii) (larger) forces between liquid molecules/(stronger) bonds B1
- (b) (i) $P_1V_1 = P_2V_2$ numerical or algebraic C1
 $6(.0) \text{ cm}^3$ A1
- (ii) temperature is constant // no gas enters/leaves // mass constant B1 [6]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE O LEVEL – October/November 2011	5054	22

- 3 (a) (i) $p_1V_1 = p_2V_2$ B1
- (ii) $2.5 \times 10^7 \times 18 = 1.0 \times 10^5 \times V_2$ C1
 4500 m^3 A1
- (b) balloon inflates higher up/bursts (if fully inflated on ground) B1
(atmospheric) pressure is less higher up/decreases with height B1
- OR
(otherwise) greater upthrust/upwards force B1
(otherwise) rises (too) high/fast B1 [5]

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21

- 3 (a) Pa or N/m^2 or cm of mercury or atmosphere(s) B1
- (b) correct points plotted at $(0.5V_0, 2P_0)$ and $(2V_0, 0.5P_0)$ B1
curve through points of decreasing gradient B1
- (c) molecules hit sides / piston B1
more molecules hit per second / hit more frequently B1
molecular impacts create large(r) **force** (upwards on piston) B1